

dense and begins to rise, while the cold fluid on top begins to sink. This sets up a pattern of convection rolls, completely analogous to the steady rotation of our waterwheel. With further increases of the Rayleigh number, the rolls become wavy and eventually chaotic.

The analogy to the waterwheel breaks down at still higher Rayleigh numbers, when turbulence develops and the convective motion becomes complex in space as well as time (Drazin and Reid 1981, Bergé et al. 1984, Manneville 1990). In contrast, the waterwheel settles into a pendulum-like pattern of reversals, turning once to the left, then back to the right, and so on indefinitely (see Example 9.5.2).

9.2 Simple Properties of the Lorenz Equations

In this section we'll follow in Lorenz's footsteps. He took the analysis as far as possible using standard techniques, but at a certain stage he found himself confronted with what seemed like a paradox. One by one he had eliminated all the known possibilities for the long-term behavior of his system: he showed that in a certain range of parameters, there could be no stable fixed points and no stable limit cycles, yet he also proved that all trajectories remain confined to a bounded region and are eventually attracted to a set of zero volume. What could that set be? And how do the trajectories move on it? As we'll see in the next section, that set is the strange attractor, and the motion on it is chaotic.

But first we want to see how Lorenz ruled out the more traditional possibilities. As Sherlock Holmes said in *The Sign of Four*, "When you have eliminated the impossible, whatever remains, however improbable, must be the truth."

The Lorenz equations are

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz.\end{aligned}\tag{1}$$

Here $\sigma, r, b > 0$ are parameters: σ is the *Prandtl number*, r is the Rayleigh number, and b has no name. (In the convection problem it is related to the aspect ratio of the rolls.)

Nonlinearity

The system (1) has only two nonlinearities, the quadratic terms xy and xz . This should remind you of the waterwheel equations (9.1.9), which had two nonlinearities, ωa_1 and ωb_1 . See Exercise 9.1.3 for the change of variables that transforms the waterwheel equations into the Lorenz equations.

Symmetry

There is an important *symmetry* in the Lorenz equations. If we replace $(x, y) \rightarrow (-x, -y)$ in (1), the equations stay the same. Hence, if $(x(t), y(t), z(t))$ is a solution, so is $(-x(t), -y(t), z(t))$. In other words, all solutions are either symmetric themselves, or have a symmetric partner.

Volume Contraction

The Lorenz system is *dissipative*: volumes in phase space contract under the flow. To see this, we must first ask: how do volumes evolve?

Let's answer the question in general, for any three-dimensional system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. Pick an arbitrary closed surface $S(t)$ of volume $V(t)$ in phase space. Think of the points on S as initial conditions for trajectories, and let them evolve for an infinitesimal time dt . Then S evolves into a new surface $S(t + dt)$; what is its volume $V(t + dt)$?

Figure 9.2.1 shows a side view of the volume.

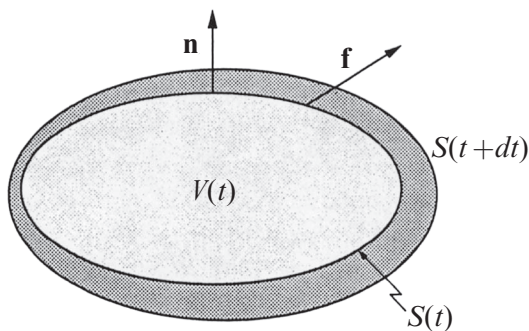


Figure 9.2.1

Let $\mathbf{n}$ denote the outward normal on S . Since $\mathbf{f}$ is the instantaneous velocity of the points, $\mathbf{f} \cdot \mathbf{n}$ is the outward normal component of velocity. Therefore in time dt a patch of area dA sweeps out a volume $(\mathbf{f} \cdot \mathbf{n} dt) dA$, as shown in Figure 9.2.2.

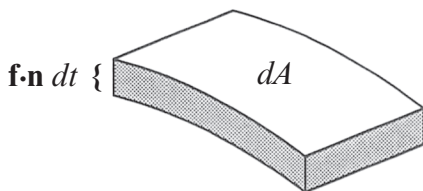


Figure 9.2.2

Hence

$$V(t + dt) = V(t) + (\text{volume swept out by tiny patches of surface, integrated over all patches}),$$

so we obtain

$$V(t + dt) = V(t) + \int_S (\mathbf{f} \cdot \mathbf{n} dt) dA.$$

Hence

$$\dot{V} = \frac{V(t + dt) - V(t)}{dt} = \int_S \mathbf{f} \cdot \mathbf{n} dA.$$

Finally, we rewrite the integral above by the divergence theorem, and get

$$\dot{V} = \int_V \nabla \cdot \mathbf{f} dV. \quad (2)$$

For the Lorenz system,

$$\begin{aligned} \nabla \cdot \mathbf{f} &= \frac{\partial}{\partial x} [\sigma(y - x)] + \frac{\partial}{\partial y} [rx - y - xz] + \frac{\partial}{\partial z} [xy - bz] \\ &= -\sigma - 1 - b < 0. \end{aligned}$$

Since the divergence is constant, (2) reduces to $\dot{V} = -(\sigma + 1 + b)V$, which has solution $V(t) = V(0)e^{-(\sigma + 1 + b)t}$. Thus *volumes in phase space shrink exponentially fast*.

Hence, if we start with an enormous solid blob of initial conditions, it eventually shrinks to a limiting set of zero volume, like a balloon with the air being sucked out of it. All trajectories starting in the blob end up somewhere in this limiting set; later we'll see it consists of fixed points, limit cycles, or for some parameter values, a strange attractor.

Volume contraction imposes strong constraints on the possible solutions of the Lorenz equations, as illustrated by the next two examples.

EXAMPLE 9.2.1:

Show that there are no quasiperiodic solutions of the Lorenz equations.

Solution: We give a proof by contradiction. If there were a quasiperiodic solution, it would have to lie on the surface of a torus, as discussed in Section 8.6, and this torus would be *invariant* under the flow. Hence the volume inside the torus would be constant in time. But this contradicts the fact that all volumes shrink exponentially fast. ■

EXAMPLE 9.2.2:

Show that it is impossible for the Lorenz system to have either repelling fixed points or repelling closed orbits. (By *repelling*, we mean that *all* trajectories starting near the fixed point or closed orbit are driven away from it.)

Solution: Repellers are incompatible with volume contraction because they are *sources* of volume, in the following sense. Suppose we encase a repeller with a closed surface of initial conditions nearby in phase space. (Specifically, pick a small sphere around a fixed point, or a thin tube around a closed orbit.) A short time later, the surface will have expanded as the corresponding trajectories are driven away. Thus the volume inside the surface would increase. This contradicts the fact that all volumes contract. ■

By process of elimination, we conclude that all fixed points must be sinks or saddles, and closed orbits (if they exist) must be stable or saddle-like. For the case of fixed points, we now verify these general conclusions explicitly.

Fixed Points

Like the waterwheel, the Lorenz system (1) has two types of fixed points. The origin $(x^*, y^*, z^*) = (0, 0, 0)$ is a fixed point for *all* values of the parameters. It is like the motionless state of the waterwheel. For $r > 1$, there is also a symmetric pair of fixed points $x^* = y^* = \pm\sqrt{b(r-1)}$, $z^* = r-1$. Lorenz called them C^+ and C^- . They represent left- or right-turning convection rolls (analogous to the steady rotations of the waterwheel). As $r \rightarrow 1^+$, C^+ and C^- coalesce with the origin in a *pitchfork* bifurcation.

Linear Stability of the Origin

The linearization at the origin is $\dot{x} = \sigma(y-x)$, $\dot{y} = rx-y$, $\dot{z} = -bz$, obtained by omitting the xy and xz nonlinearities in (1). The equation for z is decoupled and shows that $z(t) \rightarrow 0$ exponentially fast. The other two directions are governed by the system

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} -\sigma & \sigma \\ r & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

with trace $\tau = -\sigma - 1 < 0$ and determinant $\Delta = \sigma(1-r)$. If $r > 1$, the origin is a saddle point because $\Delta < 0$. Note that this is a *new type of saddle* for us, since the full system is three-dimensional. Including the decaying z -direction, the saddle has one outgoing and two incoming directions. If $r < 1$, all directions are incoming and the origin is a sink. Specifically, since $\tau^2 - 4\Delta = (\sigma+1)^2 - 4\sigma(1-r) = (\sigma-1)^2 + 4\sigma r > 0$, the origin is a stable node for $r < 1$.

Global Stability of the Origin

Actually, for $r < 1$, we can show that *every* trajectory approaches the origin as $t \rightarrow \infty$; the origin is **globally stable**. Hence there can be no limit cycles or chaos for $r < 1$.

The proof involves the construction of a **Liapunov function**, a smooth, positive definite function that decreases along trajectories. As discussed in Section 7.2, a Liapunov function is a generalization of an energy function for a classical mechanical system—in the presence of friction or other dissipation, the energy decreases monotonically. There is no systematic way to concoct Liapunov functions, but often it is wise to try expressions involving sums of squares.

Here, consider $V(x, y, z) = \frac{1}{\sigma}x^2 + y^2 + z^2$. The surfaces of constant V are concentric ellipsoids about the origin (Figure 9.2.3).

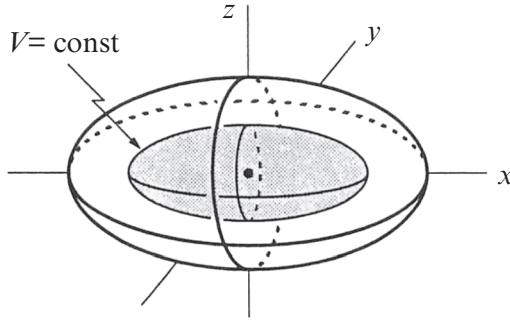


Figure 9.2.3

The idea is to show that if $r < 1$ and $(x, y, z) \neq (0, 0, 0)$, then $\dot{V} < 0$ along trajectories. This would imply that the trajectory keeps moving to lower V , and hence penetrates smaller and smaller ellipsoids as $t \rightarrow \infty$. But V is bounded below by 0, so $V(\mathbf{x}(t)) \rightarrow 0$ and hence $\mathbf{x}(t) \rightarrow \mathbf{0}$, as desired.

Now calculate:

$$\begin{aligned} \frac{1}{2}\dot{V} &= \frac{1}{\sigma}x\dot{x} + y\dot{y} + z\dot{z} \\ &= (yx - x^2) + (ryx - y^2 - xyz) + (zxy - bz^2) \\ &= (r+1)xy - x^2 - y^2 - bz^2. \end{aligned}$$

Completing the square in the first two terms gives

$$\frac{1}{2}\dot{V} = -\left[x - \frac{r+1}{2}y\right]^2 - \left[1 - \left(\frac{r+1}{2}\right)^2\right]y^2 - bz^2.$$

We claim that the right-hand side is strictly negative if $r < 1$ and $(x, y, z) \neq (0, 0, 0)$. It is certainly not positive, since it is a negative sum of squares. But could $\dot{V} = 0$?

That would require each of the terms on the right to vanish separately. Hence $y = 0$, $z = 0$, from the second two terms on the right-hand side. (Because of the assumption $r < 1$, the coefficient of y^2 is nonzero.) Thus the first term reduces to $-x^2$, which vanishes only if $x = 0$.

The upshot is that $\dot{V} = 0$ implies $(x, y, z) = (0, 0, 0)$. Otherwise $\dot{V} < 0$. Hence the claim is established, and therefore the origin is globally stable for $r < 1$.

Stability of C^+ and C^-

Now suppose $r > 1$, so that C^+ and C^- exist. The calculation of their stability is left as Exercise 9.2.1. It turns out that they are linearly stable for

$$1 < r < r_H = \frac{\sigma(\sigma + b + 3)}{\sigma - b - 1}$$

(assuming also that $\sigma - b - 1 > 0$). We use a subscript H because C^+ and C^- lose stability in a Hopf bifurcation at $r = r_H$.

What happens immediately after the bifurcation, for r slightly greater than r_H ? You might suppose that C^+ and C^- would each be surrounded by a small stable limit cycle. That would occur if the Hopf bifurcation were supercritical. But actually it's *subcritical*—the limit cycles are *unstable* and exist only for $r < r_H$. This requires a difficult calculation; see Marsden and McCracken (1976) or Drazin (1992, Q8.2 on p. 277).

Here's the intuitive picture. For $r < r_H$ the phase portrait near C^+ is shown schematically in Figure 9.2.4.

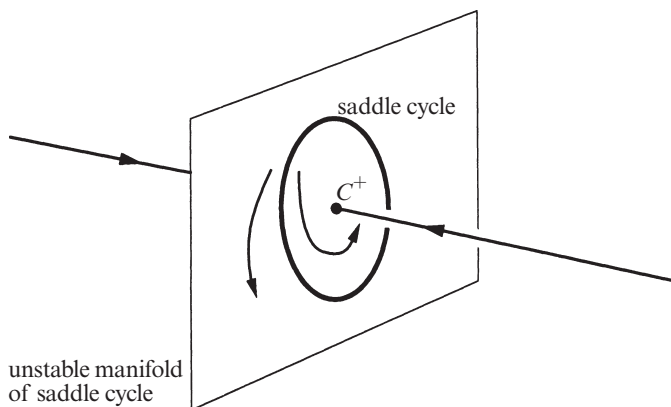


Figure 9.2.4

The fixed point is stable. It is encircled by a *saddle cycle*, a new type of unstable limit cycle that is possible only in phase spaces of three or more dimensions. The cycle has a two-dimensional unstable manifold (the sheet in Figure 9.2.4), and a two-dimensional stable manifold (not shown). As $r \rightarrow r_H$ from below, the cycle

shrinks down around the fixed point. At the Hopf bifurcation, the fixed point absorbs the saddle cycle and changes into a saddle point. For $r > r_H$ there are no attractors in the neighborhood.

So for $r > r_H$ trajectories must fly away to a distant attractor. But what can it be? A partial bifurcation diagram for the system, based on the results so far, shows no hint of any stable objects for $r > r_H$ (Figure 9.2.5).

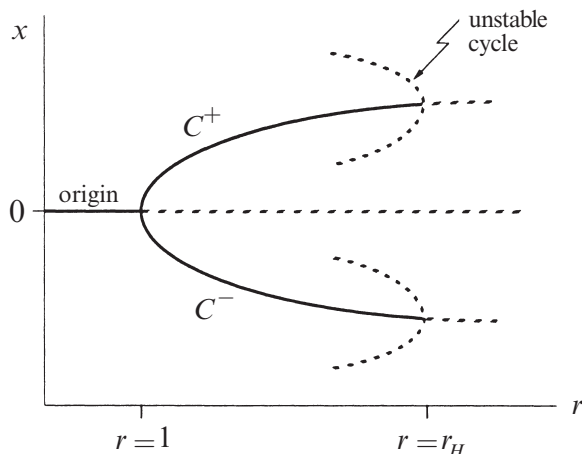


Figure 9.2.5

Could it be that all trajectories are repelled out to infinity? No; we can prove that all trajectories eventually enter and remain in a certain large ellipsoid (Exercise 9.2.2). Could there be some stable limit cycles that we're unaware of? Possibly, but Lorenz gave a persuasive argument that for r slightly greater than r_H , any limit cycles would have to be *unstable* (see Section 9.4).

So the trajectories must have a bizarre kind of long-term behavior. Like balls in a pinball machine, they are repelled from one unstable object after another. At the same time, they are confined to a bounded set of zero volume, yet they manage to move on this set forever without intersecting themselves or others.

In the next section we'll see how the trajectories get out of this conundrum.

9.3 Chaos on a Strange Attractor

Lorenz used numerical integration to see what the trajectories would do in the long run. He studied the particular case $\sigma = 10$, $b = \frac{8}{3}$, $r = 28$. This value of r is just past the Hopf bifurcation value $r_H = \sigma(\sigma + b + 3)/(\sigma - b - 1) \approx 24.74$, so he knew that something strange had to occur. Of course, strange things could occur for another reason—the electromechanical computers of those days were unreliable and difficult to use, so Lorenz had to interpret his numerical results with caution.